

The Gettysburg Address

MATH 213

Can you avoid sampling bias?

- The next slide shows Lincoln's Gettysburg Address. The entire population of words – that is, all words in his address -- will be shown to you. What is the average word length?
- **Your task: Select a sample of 10 words that resemble the overall address. Write them down.**
- **Calculate the average number of letters for the words in your sample.**
- [When you are done, enter your sample average word length here.](#) Class code is 8F34C0.

Lincoln's Gettysburg Address

“Four score and seven years ago our fathers brought forth, on this continent, a new nation, conceived in Liberty, and dedicated to the proposition that all men are created equal. Now we are engaged in a great civil war, testing whether that nation, or any nation so conceived and so dedicated, can long endure. We are met on a great battle-field of that war. We have come to dedicate a portion of that field, as a final resting place for those who here gave their lives that that nation might live. It is altogether fitting and proper that we should do this. But, in a larger sense, we can not dedicate—we can not consecrate—we can not hallow—this ground. The brave men, living and dead, who struggled here, have consecrated it, far above our poor power to add or detract. The world will little note, nor long remember what we say here, but it can never forget what they did here. It is for us the living, rather, to be dedicated here to the unfinished work which they who fought here have thus far so nobly advanced. It is rather for us to be here dedicated to the great task remaining before us—that from these honored dead we take increased devotion to that cause for which they here gave the last full measure of devotion—that we here highly resolve that these dead shall not have died in vain—that this nation, under God, shall have a new birth of freedom—and that government of the people, by the people, for the people, shall not perish from the earth.”

Random sampling

- Actual average: ???
- People are TERRIBLE at selecting a good sample, even when explicitly trying to avoid sampling bias!
- We need a better way...
- Imagine putting the names of all the units of the population into a hat, and drawing out names at random to be in the sample
- More often, we use technology

Take a **RANDOM**
sample!

Random Numbers

1. Visit www.random.org and have it select 10 numbers between 1 and 268 (the number of words in the address)
1. Using the next slide, write down the 10 words that correspond to your numbers.
1. Calculate the average number of letters in your new selection of 10 words.
1. [2 - Enter your sample average word length here.](#) Class code is F184C4.

1	Four	35in	69dedicate	103But,	137add	171here	205these	239that
2	score	36a	70a	104in	138or	172to	206honored	240this
3	and	37great	71portion	105a	139detract.	173the	207dead	241nation,
4	seven	38civil	72of	106larger	140The	174unfinished	208we	242under
5	years	39war,	73that	107sense,	141world	175work	209take	243God,
6	ago,	40testing	74field	108we	142will	176which	210increased	244shall
7	our	41whether	75as	109cannot	143little	177they	211devotion	245have
8	fathers	42that	76a	110dedicate,	144note,	178who	212to	246a
9	brought	43nation,	77final	111we	145nor	179fought	213that	247new
10	forth	44or	78resting	112cannot	146long	180here	214cause	248birth
11	upon	45any	79place	113consecrate,	147remember,	181have	215for	249of
12	this	46nation	80for	114we	148what	182thus	216which	250freedom,
13	continent	47so	81those	115cannot	149we	183far	217they	251and
14	a	48conceived	82who	116hallow	150say	184so	218gave	252that
15	new	49and	83here	117this	151here,	185nobly	219the	253government
16	nation:	50so	84gave	118ground.	152but	186advanced.	220last	254of
17	conceived	51dedicated,	85their	119The	153it	187It	221full	255the
18	in	52can	86lives	120brave	154can	188is	222measure	256people,
19	liberty,	53long	87that	121men,	155never	189rather	223of	257by
20	and	54endure.	88that	122living	156forget	190for	224devotion,	258the
21	dedicated	55We	89nation	123and	157what	191us	225that	259people,
22	to	56are	90might	124dead,	158they	192to	226we	260for
23	the	57met	91live.	125who	159did	193be	227here	261the
24	proposition	58on	92It	126struggled	160here.	194here	228highly	262people,
25	that	59a	93is	127here	161It	195dedicated	229resolve	263shall
26	all	60great	94altogether	128have	162is	196to	230that	264not
27	men	61battlefield	95fitting	129consecrated	163for	197the	231these	265perish
28	are	62of	96and	130it,	164us	198great	232dead	266from
29	created	63that	97proper	131far	165the	199task	233shall	267the
30	equal.	64war.	98that	132above	166living,	200remaining	234not	268earth.
31	Now	65We	99we	133our	167rather,	201before	235have	
32	we	66have	100should	134poor	168to	202us,	236died	
33	are	67come	101do	135power	169be	203that	237in	
34	engaged	68to	102this.	136to	170dedicated	204from	238vain,	

Sampling Bias

Sampling bias occurs when the method of selecting a sample causes the sample to differ from the population in some relevant way.

With sampling bias, certain segments of the population can become over-represented, and others under-represented.

If sampling bias exists, we cannot trust generalizations from the sample to the population

- 4.29 letters